Program overview

14-Dec-2022 11:18

Year	**2022/2023**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
TI-Mi-225	**TI-Mi-225-22 Engineering With AI**		
TI3105TU	Introduction to Python Programming	5	
TI3111TU	Algorithms and Data Structures	5	
TI3140TU	Introduction to AI and Engineering Responsible AI	5	
TI3145TU	Introduction to Machine Learning	5	
TI3150TU	Capstone Applied AI project	10	

Year	**2022/2023**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

TI-Mi-225	
Minor Coordinator	Dr. N. Yorke-Smith
Minor Coordinator	P. Pawelczak
Minor Coordinator	T.J. Viering
Intended for	The courses within this minor programme are only accessible to students who have registered for the Engineering with Ai minor programme (via the regular registration procedure).
Maximum number of participants	125
Minimum number of participants	30

TI3105TU	Introduction to Python Programming	5
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	You are expected to have finished our Python Prerequisites mini-course before this course starts. Find it at https://ipp.pages.ewi.tudelft.nl/python-prerequisites/	
Course Contents	This course teaches programming in Python on an intermediate level.	
Study Goals	After completion of this course, the student is able to: 1. Implement Python programs based on a given natural-language description. 2. Write code that follows Python's code style guidelines. 3. Make use of basic data structures such as strings, lists, dictionaries, tuples and sets. 4. Apply object-oriented programming concepts such as inheritance, operator overloading and properties. 5. Apply functional programming concepts such as list comprehensions, recursion and higher-order functions, as well as iterators and generators. 6. Implement reading and writing data from/to different file formats. 7. Understand and write type hints.	
Education Method	Seminars and lab sessions.	
Computer Use	Students do all programming assignments on their own computers (except for the exams).	
Books	Think Python 2nd Edition by Allen B. Downey (freely available from thinkpython2.com)	
Assessment	Midterm exam (2 hours) in week 1.5, final exam (3 hours) in week 1.10. Both exams are computer exams. The midterm exam covers the course materials discussed until week 1.5, and the final exam covers all course materials. There is also a resit opportunity. The resit covers all course materials (there is no separate resit for the midterm exam). The final grade of the course is calculated as follows: course grade = 0.2*max(midterm, final) + 0.8*final So the midterm grade only counts if it is higher than the grade for the final exam. The final exam grade has to be at least 5.0. (And as usual, the course grade has to be at least 5.8 in order to pass the course.) The resit grade replaces the original grade only if the resit grade is greater than the original grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You may use the Python documentation during the exam.	
Enrolment / Application	This course is only accessible to students who have registered for one of these minor programmes (via the regular registration procedure): Computer Science, Engineering with AI or Robotics.	
Co-Instructor	Prof.dr. A.E. Zaidman	

TI3111TU	Algorithms and Data Structures	5
Responsible Instructor	Dr.ir. B.H.M. Gerritsen	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Python basic programming	
Course Contents	Standard data structures (lists, trees, graphs) Standard algorithms (sorting, searching) Time and space complexity of algorithms	
Study Goals	- Understand, explain, and implement standard data structures. - Understand, explain, and implement standard algorithms. - Apply standard data structures and algorithms to solve programming tasks. - Analyze and compare algorithms with respect to their time and space complexity. - Understand and explain advanced topics in algorithm design	
Education Method	Lectures Weekly assignments (non-mandatory)	
Course Relations	Starts 2 weeks into Q1 to permit students to gain just sufficient prior Python programming knowledge. This course can only be taken as part of the Computer Science Minor or AI Minor, not as an elective.	
Assessment	Final Exam (with resit). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Calculator.	
Co-Instructor	Dr.ir. B.H.M. Gerritsen	

TI3140TU	Introduction to AI and Engineering Responsible AI	5
Responsible Instructor	O. Kudina	
Instructor	O. Kudina	
Co-responsible for assignments	T.N. Coggins	
Co-responsible for assignments	J.J.C. Maas	
Contact Hours / Week x/x/x/x	2/0/0/0 lecture; 4/0/0/0 practicum	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Introduction to AI and Responsible AI Engineering (hereafter Responsible AI for short) introduces students to the history of AI development and how it got entrenched in society, as well as to the range of ethical issues that tend to arise during the development, design and application of AI in society. It invites the students to take a critical look at the AI systems by offering a systems lens connecting their technical side with the societal and institutional ones. We will avoid vague or generalizing formulations by examining concrete applications of AI in society, e.g. synthetic media in entertainment, decision-support systems in medicine and policing, human values when AI systems are concerned, as well as touch on the political and environmental aspects of AI training and deployment. The course serves as a first step in uncovering societal issues related to AI, understanding how and why they are points of concern, and applying conceptual tools from Ethics of Technology and Responsible Engineering to mitigate and/or overcome such issues. The course will provide students with the skills of critical reflection, ethical sensitivity, and deliberation about the many issues at stake concerning AI. Thus, the course should prepare students for common problems they will face in their current studies and the capstone project, and prepare them for their future careers in the technical or humanities disciplines. This course does not require any prior knowledge in philosophy or ethics of technology. All the required reading material will be provided on Brightspace prior to the course. Detailed content: Introduction to Ethics of technology and Responsible Engineering: Technology as a sociotechnical system Values in design and Value-Sensitive Design Technological mediation of morality Technology as a Social Experiment Multicultural governance of AI systems Introduction to AI and ethics of AI: AI as a Black box Bias in training data Explainability and relatable transparency Power and politics of AI Ethics of AI in public sectors (e.g. predictive policing, facial recognition, Covid19 tracking apps, etc.)	
Study Goals	Upon completion of the course, you will be able to: explain the main philosophical differences between strong and weak AI; discuss briefly the historical development of AI; explain a range of views on the relationship between AI developments and society; identify ethical features of specific AI-based technologies; apply philosophical concepts and approaches to the current debate surrounding responsible AI; formulate a position on the ethical issues at stake in current AI applications regarding responsible design ideas; write an academic paper with a concise research question, a clear structure, a justified choice of a theoretical framework, reference to academic literature and a consistent line of argumentation; reflect on the structure and academic quality of the work of own peers in a deep and specific manner. Academic Competences The course nurtures diverse academic skills in the students to perform a holistic analysis of AI as a complex sociotechnical system. As such, it promotes disciplinary competence (i.e. engaging the technical knowledge of students with the new philosophical and ethical background); intellectual skills (i.e. analysis, synthesis, literature search, referencing, academic integrity, communication skills); temporal and social context competencies (i.e. analysis of contemporary AI case studies across different times and cultures); doing research and scientific approach competencies (i.e. throughout preparing for class exercises and the final group essay); and co-operating and communicating (i.e. class debate exercises, group-based essay and peer-review exercise). The course will thus promote an encompassing range of interconnected academic skills	
Education Method	Contact hours: lectures and tutorials, weekly quizzes based on the readings, a final paper.	
Literature and Study Materials	Study materials: Readings (available on Brightspace). Become available during the course. Lecture slides, available on Brightspace after the lecture. Quizzes that parallel the lectures material, available in that week only. Suggested literature: Powers, T.M. and Ganascia J.-M. (2020). The ethics of the ethics of AI. In Dubber, M., Pasquale, F., & Das, S. (Eds.) Oxford Handbook on AI Ethics (pp. 27-51). New York, NY: Oxford University Press. Van de Poel, I. (2018). Design for value change. Ethics and Information Technology, 1-5. Kudina, O., & Verbeek, P. P. (2019). Ethics from within: Google Glass, the Collingridge dilemma, and the mediated value of privacy. Science, Technology, & Human Values, 44(2), 291-314. Aizenberg, E. and van den Hoven, J. (2020). Designing for human rights in AI. Big Data & Society, 7(2), 1-14. van Wynsberghe, A. (2021). Sustainable AI: AI for sustainability and the sustainability of AI. AI Ethics, https://doi.org/10.1007/s43681-021-00043-6.	
Assessment	Grade calculation: Mandatory attendance in tutorials and short formative assignments (quizzes and reflective tutorial posts), Final paper (100%) Weekly quizzes: Weekly quizzes will assess the students ability to understand the reading and lecture material from one week to the next and will help to promote the retention of the study material. The surveys are mandatory but not graded. The students will be given questions on the topic for that week in a Brightspace survey, based on the mandatory readings and the lecture material for that week. The surveys are graded with a pass or fail a student will only fail if they do not answer the survey at all, or if it is apparent	

	that they have not considered the relevant material in their answers. Tutorials and lectures: Lectures will be available in a pre-recorded format on Brightspace, released in a timed weekly manner. Students are expected to attend all five tutorials on campus in person. Each tutorial will have a different AI-related case study to analyze in the form of debate, workshop, stakeholder group, etc. Final paper Students will be invited to form groups of 3-5 to reflect in a written form on an ethical aspect of AI application of their choice. In the paper, students are asked to identify a specific ethical issue raised by that technology, to assess and to make recommendations on how to develop or apply AI in a more responsible and ethical way. The assignment will proceed in the following stages throughout the course: students will first be asked to submit a paper proposal, then a draft paper, after which each student will also peer review one other paper on a similar topic. The final paper will be submitted in Week 1.10. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Artificial intelligence Ethics Philosophy
Co-Instructor	P. Pawelczak

TI3145TU	Introduction to Machine Learning	5
Responsible Instructor	T.J. Viering	
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	T.J. Viering	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	For this course, some statistics and linear algebra is necessary. To ensure IO / IDE students can successfully complete the course, it is necessary to perform self-study (15-20h) during the summer. To this end, we offer below self-study material (see also Brightspace AI minor). The detailed prior knowledge is also given below. Self-study material statistics The suggested statistics MOOC is: https://classroom.udacity.com/courses/st101 You can enrol for free. We recommend to study all material up to and including lesson 20, except lessons 5, 18 and 19. Optional lessons and programming exercises can be skipped. Understanding mathematical derivations is not necessary, but you should know how to apply the formulas and understand the concepts. Self-study material linear algebra For linear algebra, we have a set of videos to explain the necessary concepts. Vectors: https://youtu.be/OIX3iwoXglg Length of a vector / norm: https://youtu.be/SojgBN3M0oA Dot product: https://youtu.be/Qchp39dqnOA Vector and linear combinations: https://youtu.be/q8orNMQ5tpU Inner product and orthogonality: https://youtu.be/c5z9bYFqOqM Describing lines and planes: https://youtu.be/ly6seqBBpgY Matrices, matrix notation: https://www.youtube.com/watch?v=yRwQ7A6jVLk Alternative videos to further improve your understanding are: Vectors: https://www.youtube.com/watch?v=SS48Om0PX6Q Dot product and length: https://www.youtube.com/watch?v=9oh520TPa44 Distances, length, orthogonality, angles: https://www.youtube.com/watch?v=MKXahTrgVso Prior knowledge linear algebra The following topics are prior knowledge: Vectors, vector addition, vector scalar multiplication, inner product, orthogonality, computing the angle between vectors, vector equation in 1D, vector equation to define a plane in 2D, matrix notation. Prior knowledge statistics A statistics course such as CSE1210 or comparable is recommended. Usually the following book is used in Delft: Dekking, Frederik Michel, et al. A Modern Introduction to Probability and Statistics: Understanding why and how. Note that this book is freely available as a free e-book from the TU Delft library website. However, this book may not be very suited to IO students, which is why we offer the MOOC instead. The following sections in this book are prior knowledge: 2.1-2.4, 3.1-3.4, 4.1-4.3, 5.1-5.2, 5.5, 6.1, 7.1-7.2, 15.1-15.2, 15.5, 16.1 -16.2.	
Course Contents	This course teaches the basics of machine learning, including theory and practical aspects.	
Study Goals	- Apply common operations (pre-processing, plotting, etc.) to datasets using Python. - Apply the statistical concepts of supervised and unsupervised learning, and recognize their limitations. - Explain the concept of reinforcement learning. - Analyze which factors impact the performance of learning algorithms. - Apply learning algorithms to datasets using Python and Scikit-learn and evaluate their performance. - Optimize a machine learning pipeline using Python and Scikit-learn. - Apply the concept of (implicit) bias to machine learning applications.	
Education Method	Lectures and lab sessions.	
Books	We will make use of two books, which are both freely available online. 1) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. Author: Aurélien Géron. This book is freely available via the TU Delft library as an e-book. 2) An Introduction to Statistical Learning: With Applications in R (first edition). Authors: Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. This book is freely available online.	
Assessment	Exam and project. The exam (3h) takes place in week 2.10 and the final assignment is due in the same week. The project can be made in groups (pairs). The lab sessions can be handed in where each one is graded pass / fail. If enough of the handed in lab sessions receive a pass, you answer or ask one question on StackOverflow, and you hand in a self-made exam question, a bonus point (10%) can be earned for the final grade. In order to pass, students must obtain a grade greater or equal to 5.0 for both the exam and final assignment, and at least a 5.8 for the final grade after rounding. There is a resit opportunity and also a possibility to repair for the final project. Resit is only necessary for parts with a failing grade (passing grades are kept for the resit). The resit grade replaces the original grade only if the resit grade is greater than the original grade. Weighting for the final grade: Project 25% Exam 65% Bonus 10% If the student didn't earn a whole bonus point or if the bonus point lowers the grade, it will not be counted. In that case, the weights are 30% final assignment and 70% exam.	

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Enrolment / Application	This course is only accessible to students who have registered for the Engineering with AI minor program (via the regular minor registration procedure).
Tags	Artificial intelligence
Co-Instructor	Dr. N. Yorke-Smith

TI3150TU	**Capstone Applied AI project**	**10**
Responsible Instructor	P. Pawelczak	
Responsible Instructor	T.J. Viering	
Contact Hours / Week x/x/x/x	This group project can be flexibly arranged. You are expected to work around 26 hours per week (or 10 EC total) on the project (per student), you can arrange yourself when to work and where to work. You will have 2 mandatory weekly meetings of 1 hour on campus, one with a supervisor from your faculty and one with a teaching assistant. In addition, there will be a mandatory midterm presentation and final presentations on campus. There are several sessions organised on campus where you can work on the project with your group with teaching assistant support (optional), and there are a few lectures / workshops that you can attend that will support you in your group project (we aim to put the recordings online but we cannot give a hard guarantee).	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course students will apply the topics learned in all previous AI Minor courses on a group project in the area of their major. Each project will be supervised by an academic staff (PhD student, postdoc, assistant/associate/full professor) from a respective department of your major studies. Those supervisors will be responsible for the generation of the research topic of the Capstone project.	
Study Goals	- Develop an AI solution for a problem in the field of your major taking ethical and societal considerations into account - Analyze information from the literature and provided by the capstone project supervisors (i.e. the generators of the idea for the particular Capstone project) - Use the Moscow method (https://en.wikipedia.org/wiki/MoSCoW_method) to describe, refine, adjust and prioritize requirements - Justify design decisions with requirements, arguments, literature citations, or (experimental) results - Manage progress and risk during a project - Evaluate an AI solution against a list of requirements - Collaborate within a team to develop an AI solution - Present an AI solution	
Education Method	Guided group project	
Assessment	During the group project multiple deliverables are expected to be presented: - Project plan (at the start of the project) - Midterm presentation - Final presentation - Source code or other deliverables/results of the project Grading is group-based and on the basis of a rubric. We will stick to 1 grade per group, unless there was a student that performed exceptionally well or poor in the group, in that case, we may decide to deviate from the group grade and issue individual grades. Projects have NO RESIT, in case of failure of this course a redo in next academic year will be enforced. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	P. Pawelczak	
Co-Instructor	Dr. N. Yorke-Smith	

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